Mukhyansh

+91 9306372437 | mukbhateja@email.com | Github | Rohtak, Haryana | Portfolio

Education

Graphic Era Hill University Uttarakhand
Bachelor of Technology in Computer Science & Engineering 2024 – Present ***CGPA: 8.50***

Siddhartha Public School Delhi
Class XII 2022-2024 ***Score: 81.6%***

Technical Skills

Languages: C, C++, Java, JavaScript, SQL
BackEnd / Databases: Node.js, MySQL, PostgreSQL
FrontEnd: HTML, CSS
Tools: Git, GitHub, Docker, VSCode, IntelliJ, Linux, Bash
Coursework: Data Structures & Algorithms, Object Oriented Programming, Operating Systems, Multithreading

Projects

ChronOS | Resource Manager (C++, WebSockets, React,Tailwind*)*

July 2026 – Present
- Developed a Unix-like Operating System simulator with a kernel architecture in C++.
- Built CPU scheduling, synchronization helpers, and deadlock recovery mechanisms.
- Simulated process lifecycle, memory management, and concurrent execution through a Kernel-inspired architecture.
- Achieved Communication through Websockets and made the UI using React.js.

Distributed File Transfer System (*C, Compression Algorithms, VSCode, TCP Sockets)*

Jan 2026 – March 2026
- Built a distributed file transfer system with self-implemented file compressor using LZW Compression Algorithm and networking concepts.
- Structured into 3 major modules: Connectivity, Compression and File Handling.
- Successfully implemented connectivity between 2 devices on the same LAN or localhost to transfer files faster and efficiently.
- Achieved 60% reduction in file size for typical text based files before transmission.

Thread Library (*C, VSCode, Context Switching, Multithreading)* Dec 2025 – March 2026
- Implemented a Pre-emptive Thread Library in C language that provided User level Thread Management with custom TCB (Thread Control Block).
- The library itself contains 8+ core threading functionalities like signals, ucontext and setitimer() etc.
- Implemented a process scheduler from scratch following the Round Robin scheduler format.
- Achieved a 35% improvement in runtime in frequent context-switching scenarios against the POSIX pthread library.

Achievements

- Solved 450+ problems in LeetCode and CodeForces.
- 1700+ rated in contests in LeetCode.
- Led the Team in Smart India Hackathon and got selected out of 300 teams.
- Got Second place in a University Level Debate Competition (Debathon).